MAIN CONTROL PANEL KEYPAD FUNCTIONS

The control panel keypads feature paving symbols that are used for the majority of paver functions. **(Figure 4-7)** LED lights illuminate to indicate the function status. Some buttons may only control the side of the machine in which they are located, or may only function on the active console. The following pages provide detailed information for each function symbol and status lights.

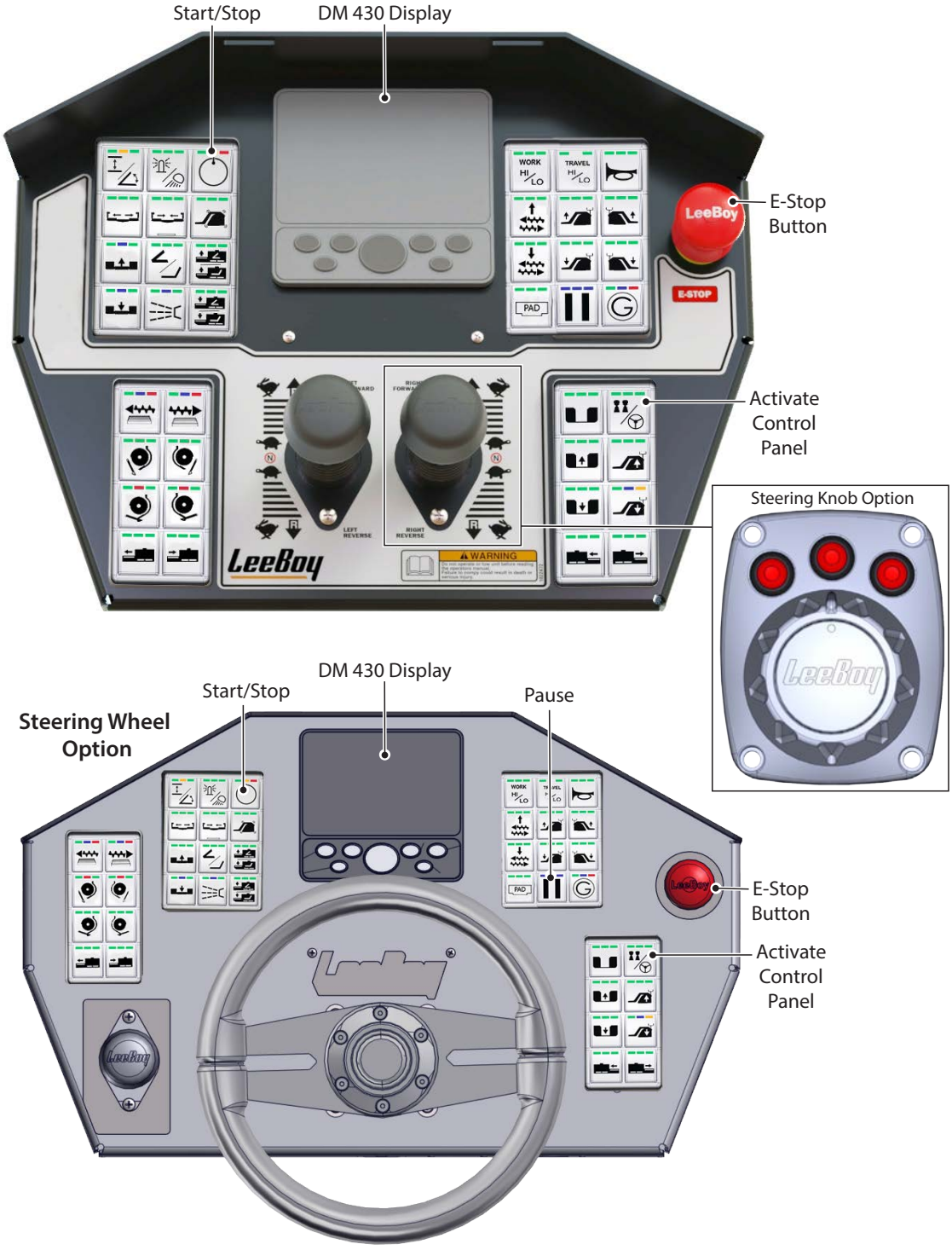


Figure 4-7. Main Control Panel

Upper Left Keypad

The upper left keypad contains push buttons for beacon/work lights, grade/slope controls, apron up/down, start, spray down, truck hitch in/out and vibrator. LED lights illuminate to indicate the function status as described below:



GRADE/SLOPE (LEFT)



The left and right tow points can be controlled automatically using the TopCon sonic automatic grade/slope control system ***(Page 4-59)*** or can be controlled manually using the keypad buttons.

- The default setting is MANUAL. The left LED light illuminates green in manual mode on both consoles. Use the Tow Up (thicker asphalt mat) and Tow Down (thinner asphalt mat) buttons to control the hydraulic tow cylinders.

NOTE: Does not function when the grade control is set to AUTO, however, the up and down buttons function when joysticks are in neutral.

- Pressing the Grade/Slope button changes to AUTO mode. The right LED light illuminates green on both consoles. The TopCon sonic control system must be connected to operate in AUTO mode.

NOTE: When in automatic mode, the Tow Up and Down buttons do not function. The tow points are paused when the machine is not traveling forward. In this situation, the center LED light will flash.



WORK/BEACON LIGHTS

Pressing this button toggles through 4 states:

- OFF (no LED lights illuminate).
- Beacon light ON (left LED light illuminates).
- Work lights ON (center LED light illuminates).
- Both beacon and work lights On (all LED lights illuminate).



START/STOP

This button is used to start and stop the engine. There are seven states that the engine start/ stop state machine can enter:

- **Boot**: Engine cannot be started while the computer system boots, but power is provided to the engine on output. The yellow and red LEDs are solid in this state. After a slight delay, the engine will leave this state and go to Ready. If the engine RPM indicates the engine is already running, then another stop cycle will be attempted (interlocks short cycling the main key switch).
- **Ready**: The yellow LED will remain on during the engine Preheat stage or if the operator attempts to start the engine while the auger, conveyor, or generator are turned on, or if one or more joysticks is not in neutral.
- **Crank**: The system will stay in the cranking state until the engine RPM indicates it is running, or the button is released. The red and green LED lights illuminate (and yellow light if still preheating). (While cranking, both the engine on and engine start outputs are active.) If the button is released before the engine is running, the system reverts back to Ready.
- **Running**: When the engine is running, only the green LED light illuminates. The engine can be stopped by pressing the Start/Stop button again.
- **Stop**: When in the stop state, the red LED light will flash for a short period of time. (The engine on output will be off in this state.) It can be held in the stop state longer if the operator presses and holds the button. The system will boot when leaving stop.
- **Shutdown**: In the case of excessively low oil pressure or high coolant temperature, a message will be sent to the engine to shut down. The red LED will flash in this state and the engine on output will remain ON. The only way to exit this state is with a power cycle.
- **Engine Offline**: When communication with the engine is lost, the system will enter an offline state. Only the yellow LED will be flashing. If the button is pressed in this state, the system will go to stop.

NOTE: All joysticks on BOTH operator main control panels *must* be in NEUTRAL to start the engine.



TRUCK HITCH or EXTENDABLE PUSH ROLLER OPTION

The truck hitch (or extendable push roller) IN/OUT function can be activated from either main control panel. All three LED lights illuminate green when pressed.

- Press and hold either button to move truck hitch (or extendable push roller) in or out as shown by arrows on the symbol.

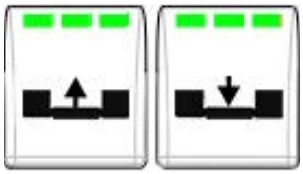


VIBRATOR

This button turns the screed vibrator ON and OFF, and can be operated from active or inactive consoles.

- Press the button once to turn on the vibrator. All LED lights illuminate green when the vibrator is engaged.
- Press the button again to turn off the vibrator.

NOTE: The vibrator only functions when traveling forward and will automatically turn off if paver is put into reverse.



APRON UP / DOWN

Press and hold the UP or DOWN buttons to pivot the front apron to assist in cleaning out the hopper. All three LED lights illuminate while in operation.



SLOPE/BERM SELECT

Use the slope/berm select button to choose either function:

- Press button once to select the Slope function.
- Press button twice to select the Berm function.
- This function only works on the activated panel as the selection is side specific.
- The left LED illuminates in slope control mode and the right LED light illuminates in berm control mode.



SLOPE/BERM UP or DOWN

After selecting slope or berm, press and hold the UP or DOWN buttons to increase/ decrease slope or berm.

- The left LED illuminates in slope control mode and the right LED light illuminates in berm control mode.
- When the button is pressed, all three LED light illuminate green on both consoles (and the corresponding screed keypad).



SPRAY DOWN

The spray down pump can be turned on and off from either main control panel whether active or inactive.

- Press the button once to turn on the spray down system. All LED lights illuminate green when the spray down system is on.

Upper Right Keypad

The upper right keypad contains push buttons for high/low gear in Work or Travel mode, horn, tow up and down functions (on each side of paver), auger lift/lower, pad builder, pause, and generator on/off. LED lights illuminate to indicate the function status as described below:





WORK HI/LO

This button controls the two speed motor shift in the WORK mode.

- Low speed is the default. The right LED light illuminates green at low speed.
- Press the button to shift to high gear. The left LED light illuminates green at high speed.
- Pressing the button again shifts back to low gear.
- To reduce downshift harshness, the operator cannot downshift at maximum speed. If a downshift is attempted at full speed, the right LED light will flash and a message banner appears on the DM430 digital display directing operator to reduce speed.

NOTE: Pave only in Travel LO (Low) mode.

NOTE: The paver will automatically move into low gear when machine operation is paused.



TRAVEL HI/LO

This button controls the two speed torque hub shift in the TRAVEL mode.

NOTE: You must come to a complete stop before shifting from LO to HI or from HI to LO gears.

- Low speed is the default. The right LED light illuminates green at low speed.
- Stop the machine and press the button to shift to high gear. The left LED light illuminates green at high speed.
- Stopping and pressing the button again shifts back to low gear.
- To reduce downshift harshness, the operator cannot downshift at maximum speed. If a downshift is attempted at full speed, the right LED light will flash and a message banner appears on the DM430 digital display directing operator to reduce speed.

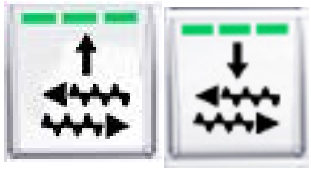
NOTE: DO NOT pave in Travel HI mode.

NOTE: The paver will automatically move into low gear when machine operation is paused.



HORN

- Pressing this button sounds the horn. All three LED lights will illuminate GREEN when engaged.
- You hold the button for longer sound.
- The horn functions whether the console is active or not.



AUGER LIFT/LOWER

These buttons allow you to lift and lower both augers.

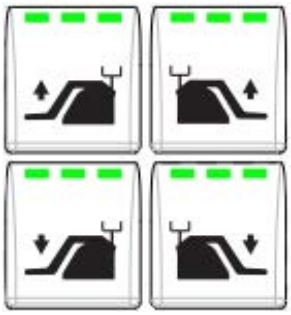
To raise the augers:

- Press and hold the auger lift button (⇧) to raise both the left and right auger.
- Releasing the button will hold the augers at that position.

To lower the augers:

- Press and hold the auger lower button (⇩) to lower both the left and right auger.
- Releasing the button will hold the augers at that position.

NOTE: ALWAYS raise the augers when loading the paver onto transport vehicles.



TOW POINTS UP or DOWN (both sides)

The left and right tow points can be controlled automatically using the TopCon sonic automatic grade/slope control system or can be controlled manually using the keypad buttons.

- The default setting is MANUAL. The left LED light illuminates green in manual mode on both consoles. Use the Tow Up (thicker asphalt mat) and Tow Down (thinner asphalt mat) buttons to control the hydraulic tow cylinders.

NOTE: Does not function when the grade control is set to AUTO, however, the up and down buttons function when joysticks are in neutral.

- Pressing the Grade/Slope button changes to AUTO mode. The right LED light illuminates green on both consoles. The TopCon sonic control system must be connected to operate in AUTO mode. (The Tow Up and Down buttons do not function in AUTO mode).

NOTE: When in automatic mode, the tow points are paused when the machine is not traveling forward. In this situation, the center LED light will flash.



PAD BUILDER

This function allows you to build a starter pad.

- Press the button once to engage the augers and conveyors. Both Under Auger Material Cut-Offs will open.
- Press the button again and the augers/conveyors will turn off.
- Press and hold button to manually engage both conveyors only.



PAUSE

When paused, the propel, augers, and conveyors will slow to a stop. All three LED lights illuminate blue when paused.

- Press the button once to pause operation. Press the button again to resume.
- If the screed temperature is above the set point, the engine will auto idle after the normal delay time.
- If the generator needs to run, the engine RPM will increase as necessary while paused.

NOTE: The engine idle will remain at high but automatically go into low idle if paused for a preset period of time (unless the generator is enabled.)

DANGER The Pause button should NEVER be used as though it's a park brake--it is not. DO NOT leave the operator platform when paused!



GENERATOR

The generator can be activated in AUTO or MANUAL mode from the active or inactive console. The default setting is AUTO mode.

- Press the button once for AUTO operation (the temperature set point controls operation). The left LED lights illuminate green.

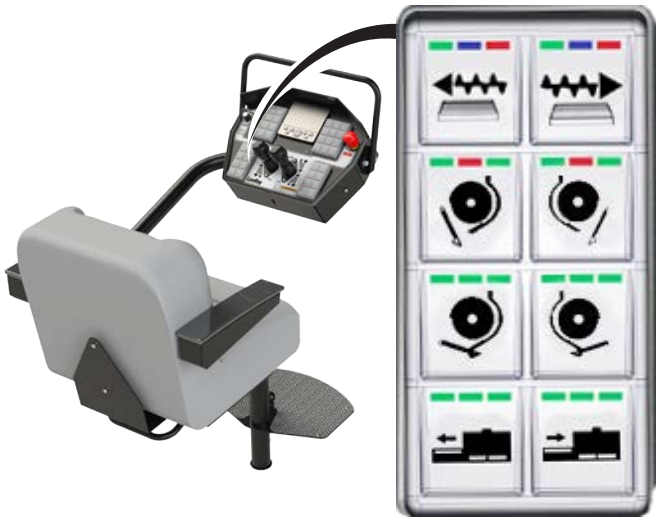
NOTE: In auto mode, if the temperature is above the set point, the center LED is solid blue. If is below the set point, the LED will flash.

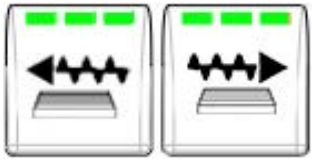
- Press the button again to turn off (the right LED light illuminates red).
- In manual mode the center LED never illuminates.

NOTE: Operator selects AUTO or manual operation in the Operator Settings screen on the DM430 display control unit. ***(Page 4-39)***

Lower Left Keypad

The lower left keypad contains push buttons for left and right auger/conveyor function, Under Auger Material Cut-off open and close, and extend or retract the left screed extension. LED lights illuminate to indicate the function status as described below:



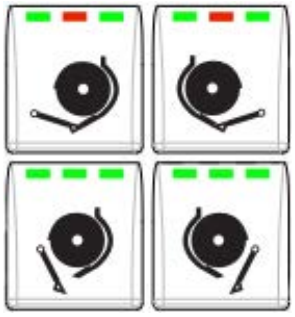


AUGERS/CONVEYORS

The augers and conveyors (both sides) can be turned on and off from either control panel. The default setting for both functions is the AUTO mode of operation.

- Press the button once to turn on the auger and conveyor on the side corresponding with the arrow (left or right side).
- The left LED illuminates green when that auger/conveyor is on; the right LED illuminates red when that auger/conveyor is off; and the center LED light illuminates blue if paver is paused while auger(s)/conveyor(s) are on (or the asphalt material has shut off the sensor).
- Press the button again to turn off the left or right conveyor.
- Pressing and holding either button is a manual override for the augers and conveyors as long as you hold the button.

NOTE: Moving the joysticks into neutral pauses auger/conveyor operation if this function is enabled.



CUT-OFFS

Both control panels and the screed keypad has buttons for controlling the left and right side Under Auger Material Cut-Offs.

- Press and hold to open or close the corresponding left or right Under Auger Material Cut-Offs.
- All three LED lights illuminate GREEN on both operator consoles and the LEFT screed keypad while open.
- If paver is put into reverse, the cut-offs will automatically close and the center LED light on both operator consoles (and the corresponding screed control keypad) illuminates RED.

NOTE: User can turn off this feature if desired.



SCREED EXTENSION (LEFT)

- Press and hold button to extend or retract the LEFT screed extension to desired width.
- Release button when desired width is achieved.
- All LED lights illuminate green when the button is pressed on the main control panel and on the corresponding left or right screed keypad.

Lower Right Keypad

The lower right keypad contains push buttons to activate the control panel consoles, operate the hopper wings, raise and float the screed, and extend or retract the right screed extension. LED lights illuminate to indicate the function status as described below:





HOPPER WING SELECT

Used to select the right, left or both hopper wings. Upon start-up, the operational mode is set by default to operating both hopper wings (the LED lights illuminate green.

- Pressing the button once enables operating only the LEFT hopper wing. The left LED light illuminates green.
- Pressing the button again operates only the RIGHT hopper wing. The right LED light illuminates green.
- Pressing the button again returns to the default operational control of both hopper wings. All three LED lights illuminate green.
- This button functions whether the console is active or not.



ACTIVATE CONSOLE

Both main control panels are inactive upon start-up. You must activate the right or left control panel to operate on the desired side of the paver. All LED lights are illuminated green on the side that is active.

- All four joysticks must be in the NEUTRAL position before either control panel can be activated.

NOTE: All joysticks on both operator platform control panels *must* be in NEUTRAL to deactivate the console in use and activate the other console.

- Press the button on the desired main control panel once to activate that control panel.
- Press the button again to deactivate an active control panel.

NOTE: If one control panel is active and the other control panel's joystick(s) is moved out of neutral, the active console will automatically deactivate.

NOTE: If an active console becomes inactive while the paver is moving, the paver will automatically slow down to a stop and apply park brake.

- A control panel cannot be activated if a parked regeneration is active. ***(Page 4-44)***



HOPPER WING OPEN

After selecting the left, right or both hopper wings on the hopper wing select button, use this button to open the selected hopper wing(s).

- Press and hold the button to open the hopper wing(s). The center LED light illuminates green while engaged.
- This button functions whether the console is active or not.



HOPPER WING CLOSE

After selecting the left, right or both hopper wings on the hopper wing select button, use this button to open the selected hopper wing(s).

- Press and hold the button to close the hopper wing(s). The center LED light illuminates green while engaged.
- This button functions whether the console is active or not.



SCREED RAISE

This function can only be operated from the active console.

- Press and hold button to raise the screed. All three LED lights illuminate GREEN while engaged (on both control panels).



SCREED FLOAT

This button controls floating the screed.

- Press the button to activate screed float (the center LED light illuminates green).
- Placing the joysticks into neutral while screed float is active moves into screed hold, and float resumes when joysticks are returned to the forward position. The center LED light illuminates blue when in screed hold.



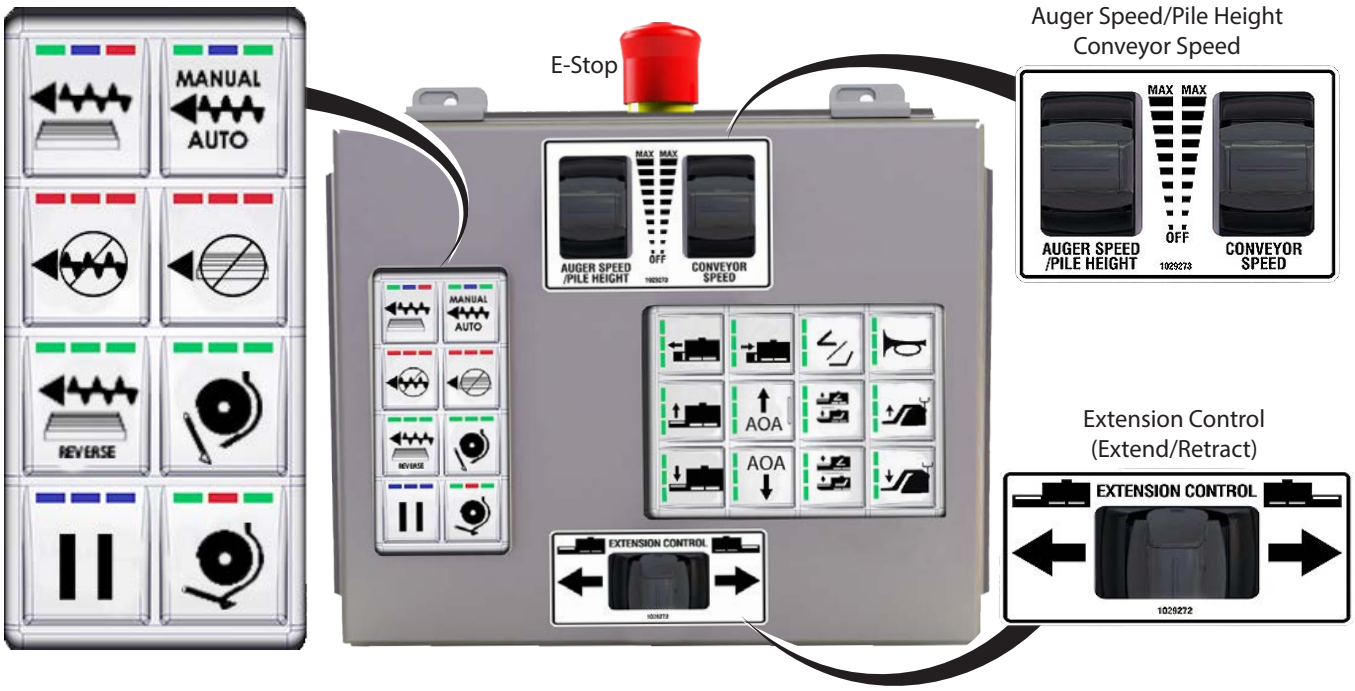
SCREED EXTENSION (RIGHT)

- Press and hold button to extend or retract the RIGHT screed extension to desired width.
- Release button when desired width is achieved.
- All LED lights illuminate green when the button is pressed on the main control panel and on the corresponding left or right screed keypad.

SCREED CONTROL BOX

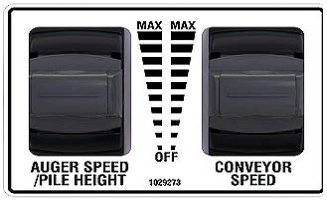
(Left Screed Control Box Shown)

Left Keypad and Control Buttons



A screed control box is located on each side of the screed, featuring keypad button controls for screed functions, auger and conveyor speed buttons, extension control buttons (extend/retract) and an emergency stop (E-Stop) button, described on the following pages.

E-Stop: Pull up on the E-Stop knob to disengage; push knob down to engage *emergency* **stopping only.**



AUGER SPEED/PILE HEIGHT

- When operating in AUTO mode, press and hold button in the desired increase/decrease direction to set the pile height at the end gate.
- When operating in MANUAL mode, press and hold button in the desired increase/decrease direction to set the auger speed.

CONVEYOR SPEED

- Press and hold the button in the desired increase/decrease direction to set conveyor speed. You can also fine tune the conveyor speed in the digital display. ***(See Motor Settings on Page 4-40)***

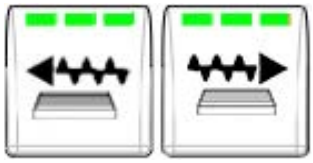


EXTENSION CONTROL

- Press and hold the button in the OUT direction to extend the screed extension.
- Press and hold the button in the IN direction to retract the screed extension.

AUGERS/CONVEYORS

The augers and conveyors (both sides) can be turned on and off from either control panel. The default setting for both functions is the AUTO mode of operation.

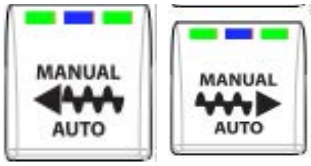


- Press the button once to turn on the auger and conveyor on the side corresponding with the arrow (left or right side).
- The left LED illuminates green when that auger/conveyor is on; the right LED illuminates red when that auger/conveyor is off; and the center LED light illuminates blue if paver is paused while auger(s)/conveyor(s) are on (or the asphalt material has shut off the sensor).
- Press the button again to turn off the left or right conveyor.

NOTE: Moving the joysticks into neutral pauses auger/conveyor operation if this function is enabled.

AUGERS MANUAL/AUTO

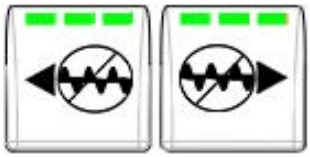
Independent right and left augers can be operated in MANUAL or AUTO (automatic) mode. Arrow direction on the symbols indicates left or right auger. The default is set for AUTO operation.



- Turn on the left or right auger by pressing the corresponding Manual/Auger button once to enable the auger on that side to operate in AUTO mode. The left LED light on the button illuminates green.
- Press and hold button 1/2 second to operate in AUTO mode. The right LED light on the button illuminates green.

NOTE: The auger sonic sensor must be connected to operate in AUTO mode. ***(Pages 4-53 to 4-54)***

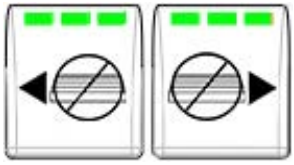
- The center blue light illuminates when material reaches the set material height or if paused.



AUGER BYPASS

If the Auger/Conveyor is on, but the augers are not needed while paving, the screed operator can press the corresponding left or right Auger Bypass button(s) to cease auger operation. All three LED illuminate to when auger operation is paused.

- Press the button again to resume auger operation.



CONVEYOR BYPASS

If the Auger/Conveyor button is on, but the conveyors are not needed while paving, the screed operator can press the corresponding left or right conveyor bypass button to cease operation. All three LED illuminate to when the conveyor operation is paused.

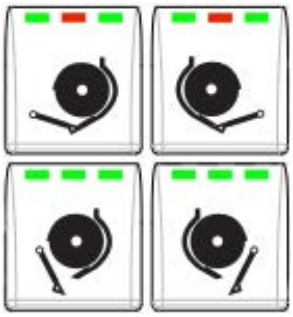
- Press the button again to resume conveyor operation.



AUGER/CONVEYOR REVERSE

Screed operators can reverse the augers and/or conveyors to help clear material as needed while paving. The default is reversing both functions. All three LED lights illuminate green in this mode.

- Press the button once to reverse only the auger(s). The left LED light will illuminate green.
- Press the button again to return to reversing both functions.



CUT-OFFS

Both control panels and the screed keypad has buttons for controlling the left and right side Under Auger Material Cut-Offs.

- Press and hold to open or close the corresponding left or right Under Auger Material Cut-Offs.
- All three LED lights illuminate GREEN on both operator consoles and the LEFT screed keypad while open.
- If paver is put into reverse, the cut-offs will automatically close and the center LED light on both operator consoles (and the corresponding screed control keypad) illuminates RED.

NOTE: User can turn off this feature if desired.



PAUSE

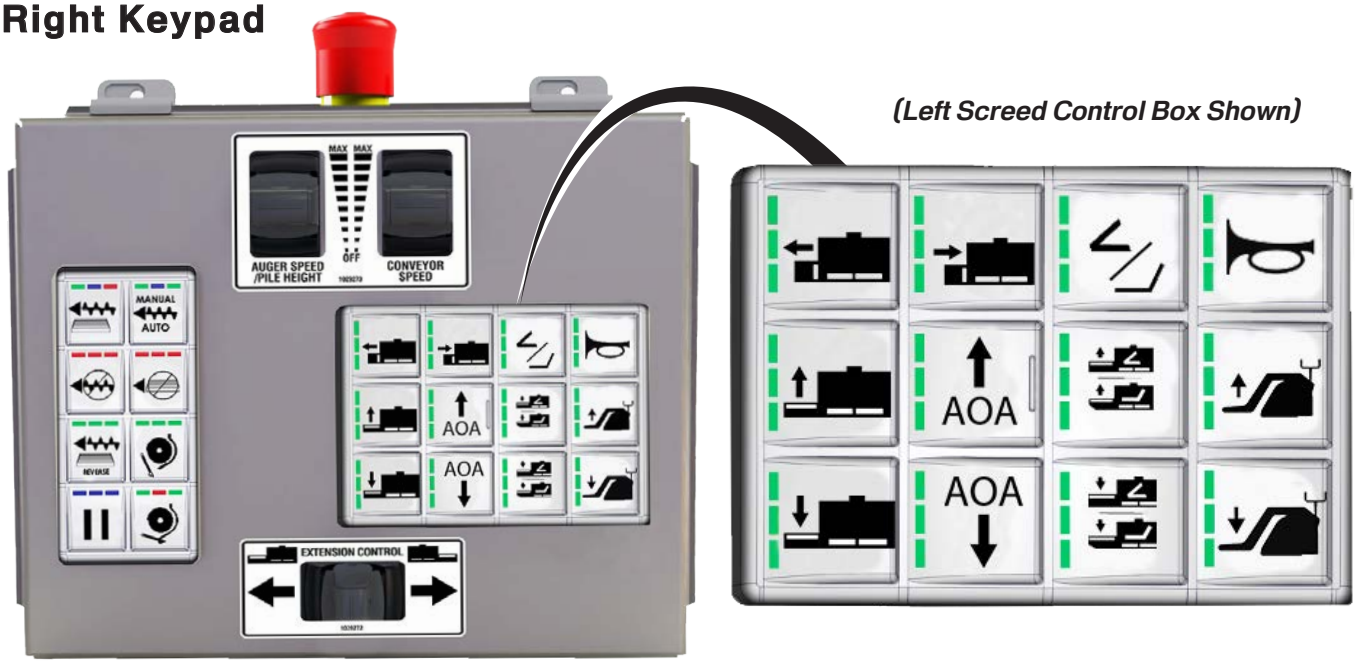
When paused, the propel, augers, and conveyors will slow to a stop. All three LED lights illuminate blue when paused.

- Press the button once to pause operation. Press the button again to resume.
- If the generator needs to run, the engine RPM will increase as necessary while paused.

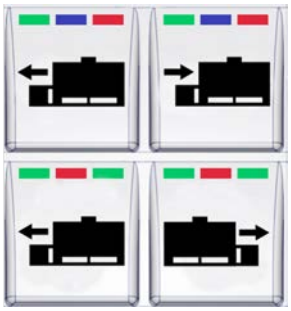
NOTE: The engine idle will remain at high but automatically go into low idle if paused for a preset period of time (unless the generator is enabled.)

DANGER The Pause button should NEVER be used as though it's a park brake--it is not. DO NOT leave the operator platform when paused!

Right Keypad



(Left Screed Control Box Shown)



TUNNEL IN/OUT

Extend and retract the tunnel when paving at wider lengths.

If operating in MANUAL mode:

- Press and hold the Tunnel Out button once to extend the tunnel. All LED lights illuminate green.
- Press and hold the Tunnel In button to retract the tunnel. All LED lights illuminate green.

If operating in AUTO mode:

- ***Simultaneously*** press and hold the extension control switch in the extend position and the Tunnel Out button to extend the tunnel.
- ***Simultaneously*** press and hold the extension control switch in the extend position and the Tunnel In button to retract the tunnel.
- To lock the tunnel in place, press and hold the Tunnel In and Tunnel Out buttons simultaneously until the LED light flashes green.
    - To unlock, press and hold both buttons again.



SLOPE/BERM SELECT

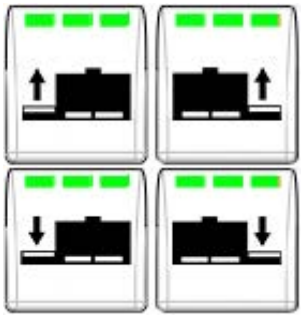
Use the slope/berm select button to choose either function. This function only works on the activated panel as the selection is side specific. The default is set at the neutral position.

- Press the button once to select the Slope function.
- Press the button twice to select the Berm function.
- The left LED illuminates in slope control mode and the right LED light illuminates in berm control mode.



HORN

- Pressing this button sounds the horn. All three LED lights will illuminate GREEN when engaged.
- You hold the button for longer sound.
- The horn functions whether the console is active or not.



VERTICAL UP/DOWN

Screed operators can use the vertical UP and DOWN buttons on the corresponding left and/or right screed control box button to raise and lower the screed extension vertical height.

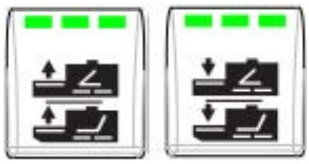
- Press and hold the corresponding button to raise or lower vertical height. All three LED illuminate while in use.



AOA UP/DOWN

The screed operator can use the AOA (Angle-of-Attack) buttons for AOA adjustments while paving.

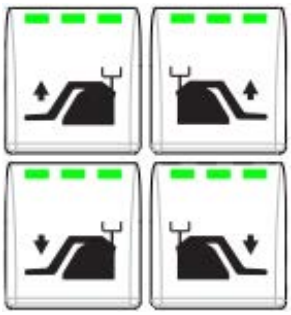
- Press and hold the UP or DOWN button to increase or decrease AOA while paving. All three LED lights illuminate while active.



SLOPE/BERM UP or DOWN

After selecting slope or berm, press and hold the UP or DOWN button to increase/ decrease slope or berm.

- The left LED illuminates in slope control mode and the right LED light illuminates in berm control mode.
- When the button is pressed, all three LED light illuminate green on both consoles (and the corresponding screed keypad).



TOW POINTS UP or DOWN (both sides)

The left and right tow points can be controlled automatically using the TopCon sonic automatic grade/slope control system or can be controlled manually using the keypad buttons.

- The default setting is MANUAL. The left LED light illuminates green in manual mode on both consoles. Use the Tow Up (thicker asphalt mat) and Tow Down (thinner asphalt mat) buttons to control the hydraulic tow cylinders.

NOTE: Does not function when the grade control is set to AUTO, however, the up and down buttons function when joysticks are in neutral.

- Pressing the Grade/Slope button changes to AUTO mode. The right LED light illuminates green on both consoles. The TopCon sonic control system must be connected to operate in AUTO mode. (The Tow Up and Down buttons do not function in AUTO mode).

NOTE: When in automatic mode, the tow points are paused when the machine is not traveling forward. In this situation, the center LED light will flash.

DPF REGENERATION (REGEN)

Before operating the paver, it is important that the operator understand the Diesel Particulate Filter (DPF) regeneration process described throughout the following pages. ***(See Section 5)*** DPF regeneration is the cleaning process where the exhaust temperature is raised to safely burn off any soot and undesired combustion gas. When the DPF filter reaches a soot saturation level between 25 - 45 percent, the onboard engine ECU (Engine Control Unit) will activate a regeneration cycle. The DPF icon on the DM430 digital controller indicates the level of soot build-up on a scale of Level 0 - 5. Regeneration will occur automatically at Level 1 - 2. DPF is further explained on the following pages.

During normal operation, the DPF is cleaned automatically. Under certain conditions, the operator may have to instruct the control system to perform the cleaning process. **It is important to perform a parked, manual regeneration BEFORE the engine reaches as high as Level 4. If the DPF reaches Level 4 - 5, non-warranty factory assistance from the engine manufacturer is required.**

Once the regeneration cycle has been activated, it is important not to shut off the engine as the regeneration cycle will fail and could possibly damage the engine. The system may attempt a second regeneration, but if it is unsuccessful, the DPF filter is blocked more than 80 percent and needs to be cleaned or replaced.

The engine speed will increase during regeneration and there may be a noticeable sound difference. Once regeneration is complete, the engine will return to normal idle speed.

NOTICE **Don't idle machine any longer than necessary as too much idling can cause soot to build up on the DPF, requiring more frequent regeneration.**

There are several factors that affect how often your DPF regenerates and the life expectancy of your DPF filter:

- Grade of engine oil.
- Grade of diesel fuel.
- Use of additives.
- Mileage.
- Wear of other components.



DM430 DIGITAL DISPLAY

The DM430 digital display features a high-resolution, anti-glare color display screen optimized with wide viewing angles in a variety of lighting conditions. There are three DM430 displays on the 8680, one on each main control panel, and the screed heat control panel. Each display is configured for the left and right operator platform controls. When starting the engine, the "boot" screen automatically displays on the DM430 digital display unit, followed by the Engine screen. .

NOTICE **Keep the controller screen clean by wiping with a soft, damp cloth. Use a mild dish washing detergent if needed. To avoid scratching or discoloring the display, DO NOT use abrasive pads, scouring powders or solvents (such as alcohol, benzene or paint thinner).**

Pressing "soft key" buttons on the control unit navigates to other screens, scrolls through information, toggles between screens and allows users to make selections as described on the following pages. A banner along the bottom of some screens correspond with the four function selection buttons (■), and may change accordingly as described on the following pages. Symbols and pop-up alert messages may also appear on the engine (and pave) screens as described throughout this section. (**Table 4-2** beginning on ***Page 4-25)***



Figure 4-8. DM 430 Button Controls

1 - Button #1

2 - ESC (Engine / Engine Diagnostics Screens)

3 - Button #2

4 - Screen Navigation

5 - Button #3

6 - Home (Pave / Pressures Screens)

7 - Button #4

ESC (Engine / Engine Diagnostics) Button: Pressing this button toggles between engine and engine diagnostics. ***(Page 4-27)***

Screen Navigation Button: Pressing this button navigates as shown in **Figure 4-9**:

- Pressing left or right navigates between all other screens (settings 1, settings 2, engine faults, and system faults).
- Pressing up or down is used on a particular screen scrolls through screen information (where prompted).
- The center of the button (↵) is used to select an item (where prompted).

Home (Pave / Pressures Screens) Button: Pressing this button toggles between the pave and pressure gauge screens. ***(Pages 4-33 and 4-41)***

If an engine fault is present, the Check Engine icon illuminates solid or flashing yellow or red, depending upon the severity of the problem.

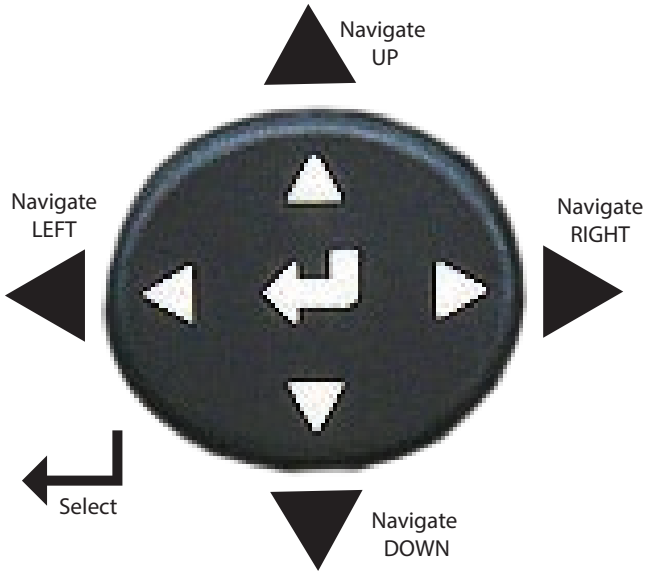


Figure 4-9. Screen Navigation Button

Engine Screen

The Engine screen is the default screen and displays various gauges. It also displays status information and icons (fuel level, DPF level, voltage, oil pressure, coolant temperature, hydraulic temperature, etc). **(Figure 4-10)**



Figure 4-10. Engine Screen

Several gauges appear on the engine screen as shown in **Figure 4-11:**

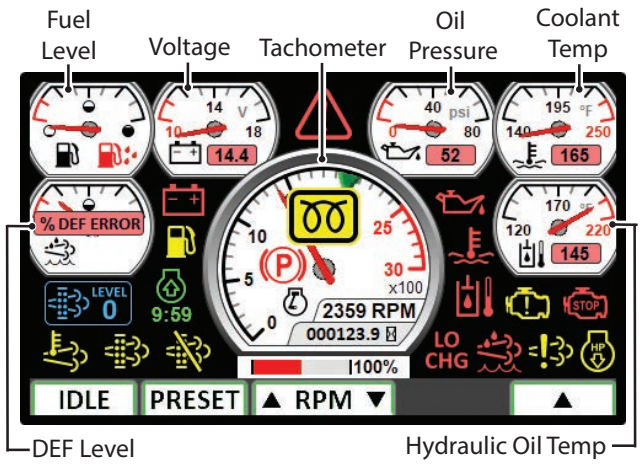


Figure 4-11. Engine Screen Gauge

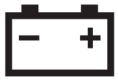
Warning symbols may also illuminate or turn red on the screen, indicating low fuel level, high exhaust temperature, low voltage, check engine, low DEF level, regeneration warnings, etc. Additional icons are shown on the screen, and will change if a fault condition is present. ***(Table 4-2 shows alert icons that may appear with descriptions.)***

- The fuel and DEF level gauges show icons to indicate fault conditions (water in fuel, improper DEF concentration).
- If the background turns red on gauges that are displayed as a digital readout, a fault exists or the computer system has lost communication.
- Engine RPM is shown in the center gauge (tachometer). The red needle is the actual RPM while the green is the RPM set point. RPMs and engine hours are also shown on the tachometer. Engine load is displayed as a red bar graph below engine hours.

DM430 Engine Icons

The following table shows icons that may appear on the engine (and other screens) to notify the operator a problem exists or component status:

Table 4-2. DM430 Engine Icons

Icon	Description
	**Park Brake.** Illuminates when the park brake is engaged.
	**Transmission Neutral.** The neutral icon displays when the transmission is in neutral. **NOTE: To perform a Parked REGEN, the "P" and "N" lamps must both be illuminated**
	**Voltage.** Illuminates if voltage level is low.
	**Oil Pressure.** Illuminates if oil pressure is too low.
	**Check Engine.** If the check engine icon illuminates, an engine problem exists that needs immediate attention. The engine may continue to operate, however, it is unable to perform DPF cleaning either automatically or manually. **⚠WARNING Take action immediately to correct the fault.**
	**Engine Stop/Protect.** This icon illuminates when an existing engine fault has been allowed to progress to such a critical level that the engine will shut down automatically to protect itself.
	**Fuel.** Illuminates when the fuel level is low.
	**Coolant Temperature.** Illuminates when the coolant temperature is too high.
	**Hydraulic Oil Temperature.** Illuminates when the hydraulic oil temperature is too high.
	**Plus +1 System Fault.** If this icon illuminates, a fault exists in the engine ECU (Plus +1).

Icon	Description
LO CHG	**Low Hydrostatic Charge Pressure.** If this icon illuminates, the hydraulic charge pressure is low.
	Engine Power De-Rate Restored Timer.
HP	**Engine Power De-Rate due to Active Fault.**
	DEF Level/Quality. Illuminates yellow if the DEF tank level or DEF concentration level is becoming low (below 15 percent). This icon will illuminate red if the DEF tank level or DEF concentration becomes critically low.
	High Exhaust Temperature. Illuminates to notify operator that the exhaust temperature is high. The icon will turn off when normal operating temperatures are reached after the REGEN cycle. ⚠WARNING **Be sure engine exhaust is away from any potentially combustible materials when this is illuminated.**
	DPF Regeneration. Illuminates when the system is performing regeneration. Any time this lamp begins flashing, the operator should increase the loading on the engine so regeneration is possible. ⚠WARNING **If increased load does not cause an automatic REGEN to occur, the operator should immediately perform a Parked, Manual REGEN.**
	The user may choose to inhibit the REGEN if conditions are too hazardous for high exhaust temperatures. When this lamp is illuminated, a REGEN cannot be performed and soot levels will continue to rise.
Level 0	⚠WARNING **Unless hazardous conditions exist, the REGEN Inhibit switch and this lamp should remain off.**
	Engine Emission Fault. Illuminates when a fault exists in the emission system. Icon will be yellow at the caution level and turn red when the fault has become critical.
	Engine Preheat Active. Illuminates before starting while the engine is preheating.

You can set engine RPMs, idle the engine, or use preset RPM set points in the Engine screen using the display buttons that correspond with the function banner along the bottom of the engine screen. **(Figure 4-12)**

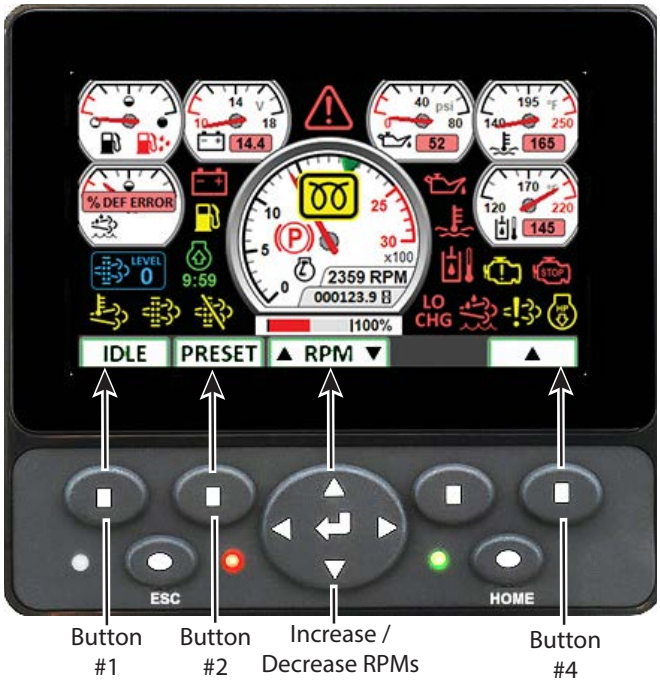


Figure 4-12. Engine Screen Navigation

- Adjust the Engine RPMs using the up (increase RPMs) and down (decrease RPMs) arrows on the center navigation button. The red needle in the tachometer gauge displays the actual RPMs.
- You can also use the preset RPM set points by pressing Button #2, which will cycle through these possible selections: Pave Low (1800 RPMs); Pave High (2100 RPMs; or Travel (2200 RPMs). The green arrow in the tachometer gauge displays set points.
- Press Button #1 to idle the engine.

NOTE: The engine will "auto idle" after a period of inactivity, and return to the previous setting when functions are activated.

The machine should be maintained in an operating condition so the DPF can automatically regenerate. When automatic regeneration occurs, the DPF regeneration icon appears on the screen. However, the operator can choose to activate the parked regeneration process, depending upon the DPF level (described on ***Pages 4-42 to 4-43).***

When you press **Button 4** in the engine screen, a pop-up banner appears with regeneration selections that correspond with the controller buttons. **(Figure 4-13)**

NOTE: If the icon background is gray, the function is unavailable (or unneeded) at the moment.

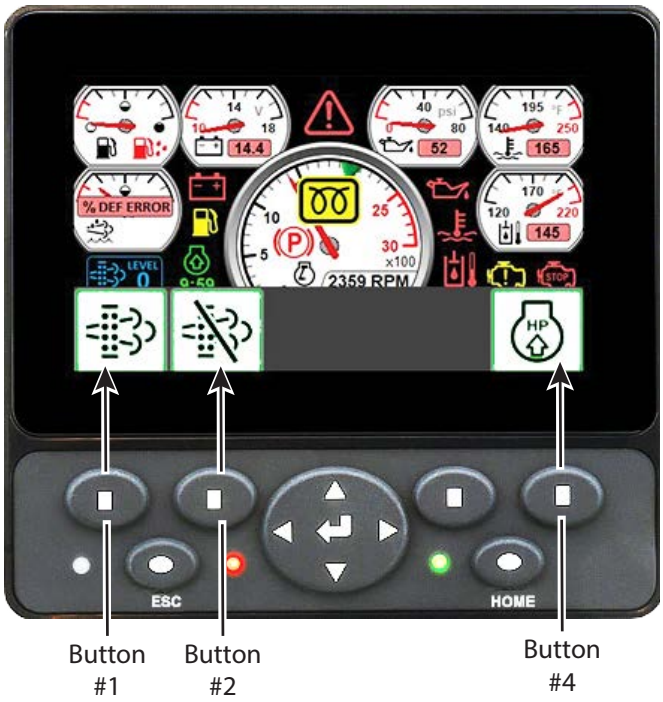


Figure 4-13. Engine Screen Pop-Up Banner

- Pressing Button #1 starts the parked regeneration process described on ***Page 4-44.***
- Pressing Button #2 inhibits the regeneration process.
- Pressing the ESC (escape) button exits the pop-up to the engine screen.

NOTE: The DEF concentration, DEF quality and DEF temperature will be shown in red when out of range.

Pressing the ESC (escape) button navigates to the Engine Diagnostics screen, which displays engine status and important information as shown in **Figure 4-14.** You can view active and logged engine fault screens *(shown on the following page)* by pressing the screen navigation button left and right. ***A complete listing of engine fault codes is shown in Trouble-shooting, Section 5.***



Figure 4-14. Engine Diagnostics Screen

- Press Button #1 to view active engine faults.
- Press Button #2 to view logged engine faults.
- Use Buttons #3 and #4 (PREV [previous] and NEXT) to scroll between fault screens.

The Active Diagnostics screen displays active warnings or faults with the appropriate Suspect Parameter Number (SPN) and Failure Mode Indicator (FMI). It may also display text description and ID/Name of the device that transmitted the message. **(Figure 4-15)**



Figure 4-15. Active Engine Faults Screen

The Logged Diagnostics screen displays nonactive warnings or faults with the appropriate Suspect Parameter Number (SPN) and Failure Mode Indicator (FMI) that are stored in the system. **(Figure 4-16)**



Figure 4-16. Logged Engine Faults Screen

Engine Alert Messages

Message banners may pop up on the engine (and pave) screen due to faults, or to alert the operator of issues. Some can be dismissed by pressing the ESC [escape] button and will periodically return to remind the operator of the issue. If the initial alert message is ignored, red alert messages will appear denoting an issue has reached critical level, and the engine output will be limited or will shut down. To prevent costly repairs, it is important to resolve a problem when it first arises.

If for some reason the DM430 controller has lost communication with the Plus +1 system or machine drive controller, the following alert message banner appears: **(Figure 4-17)**



Figure 4-17. Main Controller Offline Alert Screen

If the coolant temperature becomes too high, a red alert message banner will appear on the engine (and pave) screens to notify operator the engine will shut down. **(Figure 4-18)**

If this alert message pops up on the screen:

1. Turn off the engine (if it has not already shut down).
2. Remove key from the ignition.
3. Allow the machine to cool down several minutes, then check the coolant level. Add more coolant if needed. ***(Page 5-22)***
4. Start the engine.
5. If issues are still present or the engine will not restart, contact your authorized dealer for assistance.



Figure 4-18. High Coolant Temperature Alert Message

The following alert message appears if the engine oil pressure becomes so low that the engine will shut down. **(Figure 4-19)**

If this alert message pops up on the screen:

1. Turn off the engine (if it has not already shut down).
2. Remove key from the ignition.
3. Check the engine oil level. Add more oil if needed. ***(Page 5-17)***
4. Start the engine.
5. If issues are still present or the engine will not restart, contact your authorized dealer for assistance if needed.



Figure 4-19. Low Oil Pressure Alert Message

If the DEF is frozen, the following alert banner appears to notify operator that engine output is limited while the system attempts to thaw the DEF. You may continue operation in this state. **(Figure 4-20)**



Figure 4-20. Frozen DEF Alert Message

NOTE: Ambient temperature and humidity adversely affect DEF quality and consumption, with hot and dry climates resulting in greater DEF consumption. Temperatures above 86°F causes DEF to break down, which releases ammonia into the onboard DEF tank or storage container. DEF freezes at 12°F.

If the DEF tank temperature exceeds 120° F, the alert banner below will appear. **(Figure 4-21)** Add more DEF at your earliest opportunity.



Figure 4-21. High DEF Temperature Alert Message

If the previous warning is ignored and the DEF temperature is allowed to exceed 140° F, DEF sensor damage may occur that can adversely impact engine performance. The following URGENT message will appear: **(Figure 4-22)**



Figure 4-22. URGENT High DEF Temperature Alert Message

Other urgent alert banner messages may appear if active engine faults are ignored (or the emissions system has been tampered with), therefore becoming so severe that engine output is now limited and requires authorized dealer service. **(Figures 4-23 to 4-28)**

NOTICE **Modification or rendering inoperative any emission-related component can subject you to government penalties (and will void your warranty). Tampering with emission controls is in violation of federal law, and can result in civil penalties of up to $3,750 each day an engine or piece of equipment is operated in violation.**



Figure 4-23. URGENT Engine Output Limited Alert



Figure 4-24. Tampering Detected Alert



Figure 4-25. URGENT Engine Severely Limited Alert

If an engine fault has been resolved, the following instructional message may appear: **(Figure 4-26)**

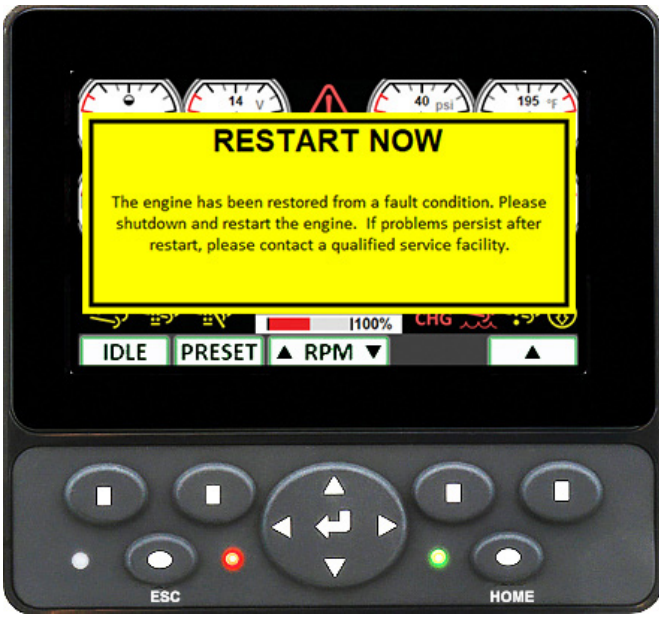


Figure 4-26. Restart Alert Message

Depending upon the engine fault, the engine output may be temporarily restored. These URGENT messages instruct the operator to move to a safe location, shut down the machine, and contact your authorized dealer for service. **(Figures 4-27 and 28)**



Figure 4-27. First Output Temporarily Restored Alert Message

If the DEF level is too low, the engine will de-rate. The restore function allows you to restore engine to full power for 10 minutes, then the engine will de-rate again if the DEF tank has not been filled. There are only two restores available per low-DEF level.

To restore engine after this event, press Button #4, which accesses the Engine screen pop-up banner.

- The engine symbol 

will be functional (not gray) in this mode. Pressing Button #4 restores.

NOTE: Restores can be used during any Level 1 torque de-rate (50 percent reduction). This can be DEF level, DEF quality, or SCR system tampering. Tampering includes any sensor or controller errors of the SCR system.

NOTICE **When the DEF level is too low, the engine will de-rate. The restore function allows you to restore engine to full power for 10 minutes, then the engine will de-rate again if the DEF tank has not been filled. There are only two restores available per low-DEF level.**



Figure 4-28. Final Output Temporarily Restored Alert Message